

Wearables and Habits: Synthetic Control Estimates

Issue #9

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1 Summary

Exploratory unit-level synthetic control of first wearable purchase on running-related purchases, as an alternative to the staggered DiD of #7 (which fails on pre-trends). **Balanced panel only**, and each adopter’s synthetic twin is matched on outcome history, purchase intensity, category mix and survey demographics (Section 2).

- **Running-related purchasing rises modestly, but so does everything else.** Post ATT (quarters 1– L): running-related +0.046 (0.017) with $L = 8$ (113 adopters) and +0.039 (0.013) with $L = 6$ (242); running shoes +0.016 (0.012) and +0.013 (0.008); supplements +0.066 (0.025) and +0.050 (0.017); fitness gear and medication are within about 1.5 SE of zero.
- **Non-running outcomes move too.** Books +0.042 (0.023) / +0.040 (0.016), grocery +0.040 / +0.010, pet +0.035 / +0.020, and total purchases $\ln(1+n)$ +0.264 (0.054) / +0.205 (0.040). Adopters buy more overall than their twins after adoption, and the running-related gain is of the same size as the books placebo, so it cannot be attributed to running specifically.
- **Pre-fit and balance.** Pre-period gaps are about zero (Figures 1, 2), so the pre-trend problem of #7 is absorbed by the twin. Category shares and most demographics are close to the adopters’ (Table 4); the exceptions are pre-period $\ln$ spend (8.22 vs 8.07 synthetic) and income band (2.89 vs 2.72).
- **Randomization p-values** (Table 3; two-sided). $L = 8$: running-related $p = 0.002$, supplements < 0.001 , running shoe 0.061, fitness gear 0.20, medication 0.082; but books 0.037, grocery 0.076, pet 0.096 and $\ln(1+n)$ $p < 0.001$. $L = 6$: running-related < 0.001 , supplements < 0.001 , running shoe 0.018, fitness gear 0.017, medication 0.61; books 0.006, grocery 0.53, pet 0.15, $\ln(1+n)$ < 0.001 . Running-related and supplements are clearly distinguishable from the placebo distribution, but so are total purchasing and books, so the test does not isolate a running effect.
- **Last pre-quarter imbalance and augmented SC.** For $\ln(1+n)$ the plain synthetic twin misses the quarter before adoption (gap at $e = -1$: +0.126 with $L = 8$, +0.105 with $L = 6$; other pre-quarters 0.01–0.04): adopters’ purchasing ramps up from about four quarters before the device (mean purchases 23, 29, 34, 41 per quarter at $e = -8, -4, -2, -1$ against a gradual drift among donors), a ramp that is not the device itself, so it is consistent with a time-varying shock (e.g. income) that donors’ convex combinations cannot reach. An augmented SC with a ridge bias correction (Section 4) removes it ($e = -1$ gap +0.011). The post effects shrink somewhat but keep their pattern: $L = 8$ running-related +0.036 (0.017, $p = 0.013$), supplements +0.058 ($p = 0.003$), $\ln(1+n)$ +0.218 ($p < 0.001$), books +0.033 ($p = 0.12$); $L = 6$ running-related +0.031 ($p < 0.001$), supplements +0.043, books +0.032 ($p = 0.034$), $\ln(1+n)$ +0.166.

- **Pseudo-adopter placebo** (random never-treated users given real adoption dates) is about zero for nearly all outcomes (largest: pet +0.034 at $L = 8$, supplements -0.027 at $L = 6$).
- **Relation to earlier runs and #7.** A preliminary unbalanced-panel run (outcome-only matching, 288 adopters; kept in `synth_summary*.csv`) gave larger gaps (running-related +0.070). The DiD of #7 gave small *negative* estimates, so the methods still disagree in sign. A plausible common driver is selection of adopters on a purchase in $e = 0$ and on ≥ 12 months of later activity, which donors are not selected on; this predicts positive gaps for all outcomes. Not tested.

2 Design

- **Balanced sample:** “clean” sample users observed in every quarter 2018Q1–2022Q4 (2,764 users). Adopters must have the full window $g - L, \dots, g + L$ inside 2018Q1–2022Q4: $L = 8$ gives adoption in 2020 (113 adopters), $L = 6$ in 2019Q3–2021Q2 (242). Donors: balanced never-treated users (2,392). No unobserved quarters, so donors are never dropped in the post-period.
- **What is matched** (donors and adopter use the same calendar pre-window $g - L, \dots, g - 1$), four blocks, each column standardized by the donor-pool SD and scaled so a block’s total weight is:
 1. weight 2: the studied outcome (any purchase in the quarter) in each pre-quarter;
 2. weight 1: $\ln(1 + \text{purchases})$ in each pre-quarter (omitted when it is the outcome);
 3. weight 1: pre-window category mix, the share of purchases in running-related, fitness gear, supplements, medication, pet, books, gift cards, cables/batteries, cleaning and grocery, plus $\ln(1 + \text{totalspend})$;
 4. weight 1: survey demographics: age band, income band, household size, female, college, white, diabetes, smoker, alcohol, and a survey-missing flag (mean-imputed as in #7).
- **Weights:** non-negative, sum to one, least squares over the 150 nearest donors (distance on the same standardized predictors). With four blocks the fit is a compromise, so pre-RMSE on the outcome is larger than with outcome-only matching.
- **Effects:** gap = adopter minus synthetic, $e = -L, \dots, L$ ($e = 0$ the adoption quarter); ATT_e = mean over adopters; SE = cross-adopter SD/ $\sqrt{n}$ (ignores weight estimation error and donor reuse, so likely too small; no bootstrap is used). **Inference:** randomization p-values, below. Post ATT = mean over $e = 1, \dots, L$. Placebo: as many never-treated users as adopters get randomly drawn real adoption quarters and are run against the remaining donors.
- Code: `code/scripts/synth_control_balanced.py` (argument L) and `synth_control_balanced_figs.py`.

3 Results

Randomization inference

Statistic: pooled post ATT. Null distribution: N never-treated users (same number as adopters) are drawn at random, given adoption quarters drawn from the real adopters’ cohort distribution, and run through the identical procedure against all other donors; 5,000 such draws, using a random pool of 500 donors $\times$ each real cohort (`synth_control_perm.py`, `synth_control_perm_tab.py`). $p = (1 + \#\{|\text{ATT}_{\text{perm}}| \geq |\text{ATT}_{\text{obs}}|\})/(R + 1)$, two-sided; this is a Monte Carlo approximation to the exact permutation p-value (full enumeration is infeasible), valid under exchangeability of adopters

Table 1: Balanced synthetic control, $L = 8$ (113 adopters). Pre RMSE = RMS adopter-minus-synthetic gap in the pre-period / RMS outcome level.

Outcome	Post ATT (SE)	Placebo (SE)	N	Pre RMSE / level
Running shoe	0.016 (0.012)	0.001 (0.005)	113	0.017 / 0.102
Running-related (keyword)	0.046 (0.017)	0.001 (0.014)	113	0.050 / 0.278
Fitness gear	0.015 (0.013)	0.004 (0.013)	113	0.027 / 0.190
Supplements	0.066 (0.025)	0.017 (0.022)	113	0.079 / 0.366
Medication	0.034 (0.020)	0.011 (0.018)	113	0.034 / 0.208
PLACEBO: pet	0.035 (0.022)	0.034 (0.023)	113	0.074 / 0.401
PLACEBO: books	0.042 (0.023)	-0.001 (0.021)	113	0.111 / 0.588
PLACEBO: grocery/food	0.040 (0.022)	0.011 (0.023)	113	0.050 / 0.266
General: $\ln(1+\#\text{purchases})$	0.264 (0.054)	-0.003 (0.062)	113	0.280 / 2.939

Table 2: Balanced synthetic control, $L = 6$ (242 adopters).

Outcome	Post ATT (SE)	Placebo (SE)	N	Pre RMSE / level
Running shoe	0.013 (0.008)	-0.009 (0.005)	242	0.007 / 0.080
Running-related (keyword)	0.039 (0.013)	0.000 (0.011)	242	0.030 / 0.253
Fitness gear	0.019 (0.010)	0.010 (0.009)	242	0.017 / 0.174
Supplements	0.050 (0.017)	-0.027 (0.015)	242	0.053 / 0.365
Medication	-0.007 (0.013)	0.006 (0.014)	242	0.023 / 0.212
PLACEBO: pet	0.020 (0.015)	-0.005 (0.018)	242	0.048 / 0.378
PLACEBO: books	0.040 (0.016)	0.015 (0.016)	242	0.087 / 0.570
PLACEBO: grocery/food	0.010 (0.017)	0.020 (0.017)	242	0.035 / 0.274
General: $\ln(1+\#\text{purchases})$	0.205 (0.040)	0.008 (0.043)	242	0.230 / 3.016

and donors, which the adoption selection discussed above may violate. The smallest attainable value is about 0.0002. The observed ATTs reproduce the tables above.

4 Augmented synthetic control

The plain estimator restricts weights to be non-negative and sum to one, so the synthetic twin cannot exceed the best donors. Adopters' purchasing rises steeply in the quarter before adoption and lies outside the donors' range, so the last pre-quarter is not matched. The augmented estimator (Ben-Michael, Feller and Rothstein 2021) adds a ridge-regression correction: for each event time e , $m_e(x)$ is a ridge fit of the outcome on the same standardized predictors over *all* donors, and the twin's prediction becomes $\sum_j w_j y_{j,e} + [m_e(x_i) - \sum_j w_j m_e(x_j)]$, which lets weights extrapolate. The ridge penalty is chosen per outcome and cohort by leave-one-donor-out error of the mean post outcome (grid 10^{-1} to 10^4). Pseudo-adopters are removed from both the shortlist and the ridge fit. The randomization inference is the same as above. Code: `code/scripts/synth_control_aug.py`, `synth_control_aug_figs.py`.

The $\ln(1+n)$ pre-period RMSE falls from 0.31 to 0.05 ($L = 8$) and the $e = -1$ gap from 0.126 to 0.011; the pre-period event-study is flat (Figure 3). Caution: the pre-period outcomes are themselves predictors in the ridge, so near-zero pre-gaps are partly mechanical and not independent evidence that the design is valid. The remaining concern is the post-period one: whether adopters' extra purchasing after adoption is caused by the device or by the same unobserved shock that preceded it.

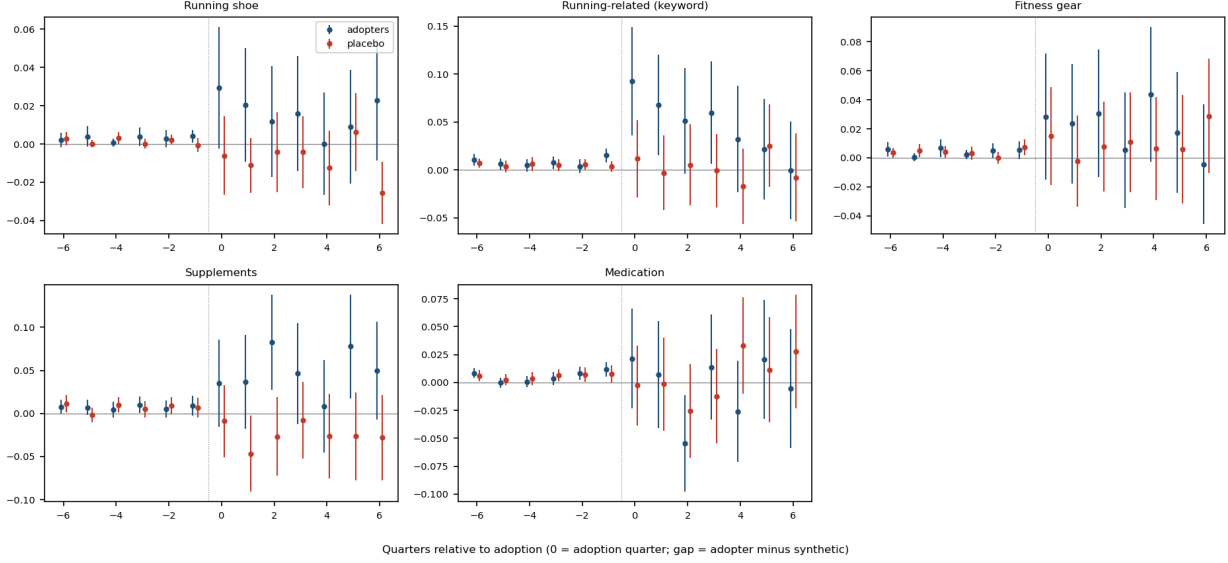


Figure 1: Event-study gaps, main outcomes, $L = 6$ (blue adopters, red pseudo-adopter placebo; 95% CIs). $L = 8$: synth_bal_es_main_L8.png.

Table 3: Randomization inference: observed ATT, SD of the permutation distribution, two-sided p.

Outcome	L=8 ATT	L=8 perm. SD	L=8 p	L=6 ATT	L=6 perm. SD	L=6 p
Running shoe	0.016	0.007	0.061	0.013	0.005	0.018
Running-related (keyword)	0.046	0.014	0.002	0.039	0.009	<0.001
Fitness gear	0.015	0.011	0.203	0.019	0.008	0.017
Supplements	0.066	0.019	<0.001	0.050	0.013	<0.001
Medication	0.034	0.016	0.082	-0.007	0.011	0.613
PLACEBO: pet	0.035	0.021	0.096	0.020	0.014	0.154
PLACEBO: books	0.042	0.020	0.037	0.040	0.014	0.006
PLACEBO: grocery/food	0.040	0.021	0.076	0.010	0.013	0.530
General: $\ln(1+\#\text{purchases})$	0.264	0.059	<0.001	0.205	0.039	<0.001

5 Caveats and follow-ups

- Balanced sample means long-lived, early-start users and adopters in a narrow calendar window (2020 for $L = 8$), so results are not comparable to #7's 447.
- Follow-ups (not done): hold-out pre-period fit; synthetic DiD; pseudo-adopters selected on an $e = 0$ purchase; not-yet-treated donors; tuning predictor block weights.

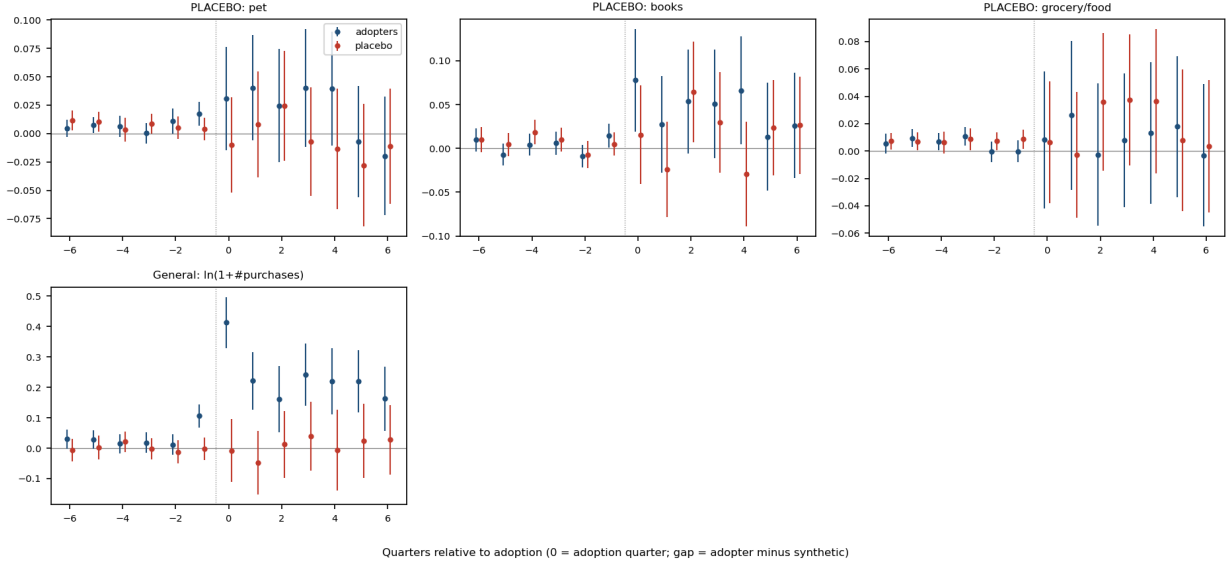


Figure 2: Same, placebo and general-intensity outcomes.

Table 4: Covariate balance, $L = 8$, running-related fit: average adopter, synthetic twin and raw donor pool.

Variable	Adopters	Synthetic	Donor pool
share: run_any	0.009	0.009	0.006
share: fit_gear	0.004	0.004	0.004
share: supp	0.017	0.018	0.019
share: med	0.006	0.006	0.006
share: pet	0.028	0.027	0.023
share: book	0.066	0.067	0.071
share: giftcard	0.015	0.019	0.024
share: cable	0.024	0.024	0.023
share: cleaning	0.004	0.004	0.004
share: grocery	0.012	0.015	0.014
ln spend (pre)	8.216	8.069	7.500
age	1.841	1.950	1.888
inc	2.885	2.720	2.321
hh	2.619	2.547	2.399
female	0.513	0.534	0.531
college	0.743	0.715	0.659
white	0.867	0.863	0.801
diab	0.081	0.096	0.126
smoke	0.090	0.104	0.127
alc	0.438	0.470	0.460
svy_miss	0.009	0.005	0.018

Table 5: Plain (SC) vs augmented synthetic control, $L = 8$ (113 adopters). p = two-sided randomization p-value; $e=-1$ = average gap in the last pre-quarter; Pre RMSE over $e = -8, \dots, -1$.

Outcome	SC ATT	SC p	Aug. ATT (SE)	Aug. p	$e=-1$ SC	$e=-1$ Aug.	Pre RMSE SC	Pre RMSE Aug.
Running shoe	0.016	0.061	0.013 (0.012)	0.118	0.012	0.002	0.055	0.009
Running-related (keyword)	0.046	0.002	0.036 (0.017)	0.013	0.024	0.002	0.081	0.009
Fitness gear	0.015	0.203	0.009 (0.013)	0.404	0.012	0.001	0.054	0.009
Supplements	0.066	<0.001	0.058 (0.025)	0.003	0.008	-0.001	0.102	0.018
Medication	0.034	0.082	0.026 (0.020)	0.139	0.019	0.002	0.061	0.012
PLACEBO: pet	0.035	0.096	0.023 (0.022)	0.266	0.030	0.003	0.099	0.025
PLACEBO: books	0.042	0.037	0.033 (0.022)	0.116	0.003	-0.002	0.128	0.018
PLACEBO: grocery/food	0.040	0.076	0.034 (0.022)	0.114	-0.008	-0.004	0.079	0.016
General: $\ln(1+\text{\#purchases})$	0.264	<0.001	0.218 (0.051)	<0.001	0.126	0.011	0.313	0.046

Table 6: Same, $L = 6$ (242 adopters).

Outcome	SC ATT	SC p	Aug. ATT (SE)	Aug. p	$e=-1$ SC	$e=-1$ Aug.	Pre RMSE SC	Pre RMSE Aug.
Running shoe	0.013	0.018	0.012 (0.008)	0.030	0.004	0.000	0.033	0.009
Running-related (keyword)	0.039	<0.001	0.031 (0.013)	<0.001	0.015	0.001	0.053	0.007
Fitness gear	0.019	0.017	0.015 (0.010)	0.053	0.005	0.000	0.039	0.005
Supplements	0.050	<0.001	0.043 (0.017)	0.003	0.009	0.000	0.075	0.014
Medication	-0.007	0.613	-0.015 (0.013)	0.257	0.012	0.001	0.044	0.007
PLACEBO: pet	0.020	0.154	0.012 (0.015)	0.419	0.018	0.002	0.073	0.015
PLACEBO: books	0.040	0.006	0.032 (0.016)	0.034	0.014	0.000	0.104	0.014
PLACEBO: grocery/food	0.010	0.530	0.003 (0.017)	0.813	-0.000	-0.001	0.057	0.010
General: $\ln(1+\text{\#purchases})$	0.205	<0.001	0.166 (0.039)	<0.001	0.105	0.008	0.269	0.031

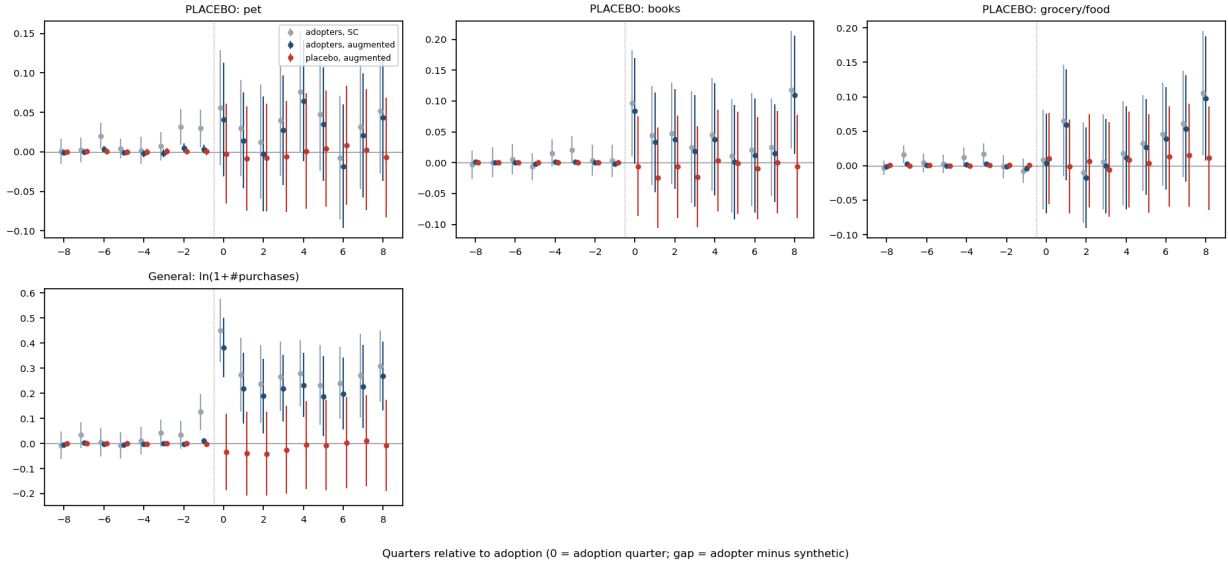


Figure 3: $L = 8$ event-study gaps for placebo and general-intensity outcomes: plain SC (grey), augmented (blue), pseudo-adopters with augmented (red). Main outcomes: `synth_aug_es_main_L8.png`.